

Quantum Computing

Basic Notions



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Physics and Computation

- **Classical Physics** can be seen as an approximation of **Quantum Physics**, i.e. the fundamental theory explaining Nature in the most accurate ways.
- **Classical Computation** can be seen as an approximation of **Quantum Computation**, i.e. a computation that makes full use of the laws of quantum mechanics.
- The theory of computation is conceptually strongly related to physics.

Computation	Physics
Computer	Physical System
Computation	Motion
Input	Initial state
Rules	Laws of motion
Output	Final state

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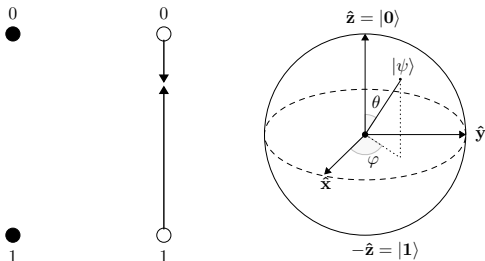


Figure: Possible configuration of a deterministic bit (left), probabilistic bit (middle) and quantum bit (right). $|\psi\rangle \equiv \cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle$, where θ and φ are the (real) **spherical coordinates**.

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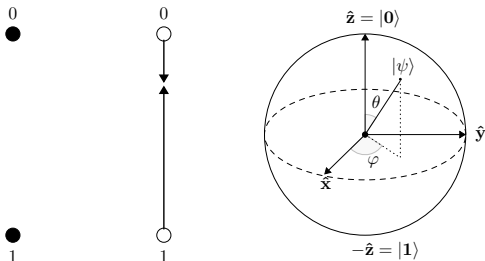


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Bloch Sphere

Representation of qubit and operations on it.

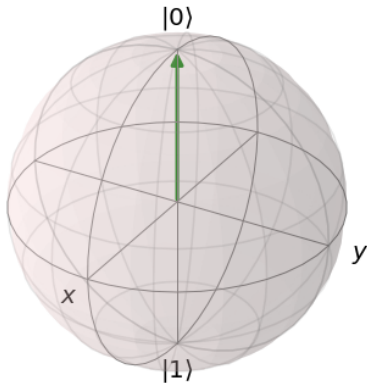


Figure: State $|0\rangle$ on the Bloch sphere

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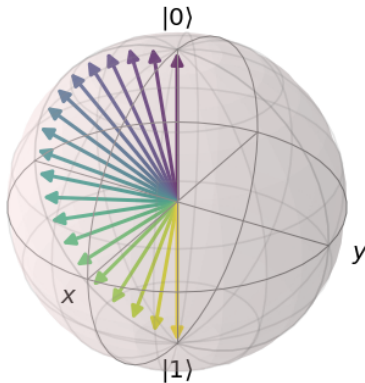


Figure: State evolving from $|0\rangle$ to $|1\rangle$ through $RX(\pi/2)$

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All unitary 2×2 complex matrices are **operations** on one qubit! (only four operations on one classical bit).

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Phase Gate $\Phi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}$

Quantum Registers



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Tensor Product (vectors)

For vectors $(x_1, x_2)^t, (y_1, y_2)^t \in \mathbb{C}^2$, the tensor product $(x_1, x_2)^t \otimes (y_1, y_2)^t$ is the vector $(x_1 y_1, x_2 y_1, x_1 y_2, x_2 y_2)^t \in \mathbb{C}^4$.

For two qubits $|a\rangle, |b\rangle$, the tensor product $|a\rangle \otimes |b\rangle = |a, b\rangle$ is a two-qubit register, with computational base:

$$\{|0\rangle, |1\rangle\} \otimes \{|0\rangle, |1\rangle\} = \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$$

In algebraic notations:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Generic two-qubit state: $|\psi\rangle = \alpha_{00} |00\rangle + \alpha_{01} |01\rangle + \alpha_{10} |10\rangle + \alpha_{11} |11\rangle$.

Tensor Product (Matrices)

For the matrices

$$M = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \text{ and } N = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} \text{ we have}$$

$$M \otimes N = \begin{bmatrix} 0 & 1 & 0 & 3 \\ -1 & 2 & -3 & 6 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 2 \end{bmatrix}.$$

If $U : \mathbb{C}^2 \mapsto \mathbb{C}^2$ is an operation on 1 qubit, then $U : \mathbb{C}^{2^n} \mapsto \mathbb{C}^{2^n}$ is an operation on n qubit registers.

Hadamard

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$(H \otimes H)|00\rangle = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}$$

Entanglement

Not all n -qubit state can be expressed as tensor product of the states of the individual qubits, i.e. (for a 2-qubit register)

$$(a_1 |0\rangle + b_1 |1\rangle) \otimes (a_2 |0\rangle + b_2 |1\rangle).$$

Bell States

The Bell or EPR states are *entangled* states:

$$\beta_{00} = (|00\rangle + |11\rangle) / \sqrt{2}$$

$$\beta_{10} = (|01\rangle + |10\rangle) / \sqrt{2}$$

$$\beta_{01} = (|00\rangle - |11\rangle) / \sqrt{2}$$

$$\beta_{11} = (|01\rangle - |10\rangle) / \sqrt{2}$$

Measurement

As for one qubit, the measurement of a n -qubit register results in one of the 2^n classical n -bit sequence with a probability given by the associated probability.

Example

Measuring in the computational base the state

$$|\psi\rangle = \alpha_{00} |00\rangle + \alpha_{01} |01\rangle + \alpha_{10} |10\rangle + \alpha_{11} |11\rangle \text{ with } \sum_{i \in \{0,1\}^2} |\alpha_i|^2 = 1$$

results in $i \in \{0, 1\}^2$ with probability $|\alpha_i|^2$.

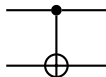
Spooky effect

When one particle of an entangled state is measured, then at the exact moment of observation, even if the particles are separated by light-years of space, the other particle takes on the state of its twin.

The CNOT operation

The most important 2-qubits operation is the CNOT (controlled-NOT) gate, with matrix representation in the standard basis $\{|0\rangle, |1\rangle\}$ and circuit diagram:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$



The circuit transforms the input $|control\rangle |target\rangle$ as follows:

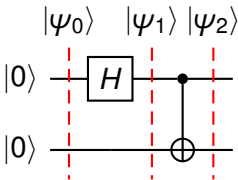
$$|00\rangle \mapsto |00\rangle, |01\rangle \mapsto |01\rangle, |10\rangle \mapsto |11\rangle, |11\rangle \mapsto |10\rangle.$$

CNOT is equivalent to the that is the classical XOR transformation

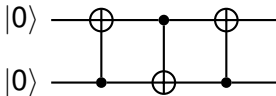
$$|A, B\rangle \mapsto |A, B \oplus A\rangle.$$

Simple Quantum Circuits

Entangling circuit



Swap circuit



Quantum Mechanics Postulates

- The state of a closed (isolated) system is described by a unit vector in a Hilbert space.
- The evolution of a system is described by an unitary operator U . Given the system in state $|\psi_1\rangle$ then after the evolution the system will be in state $|\psi_2\rangle = U|\psi_1\rangle$.
- The state space of a composite physical system is the tensor product of the state spaces of the component physical systems.
- A *quantum measurement* is a collection of operators $\{M_1, \dots, M_n\}$ where $1, \dots, n$ are the possible outcomes of the experiment.

More in the next lecture!